Six faces.
Still not one of five.

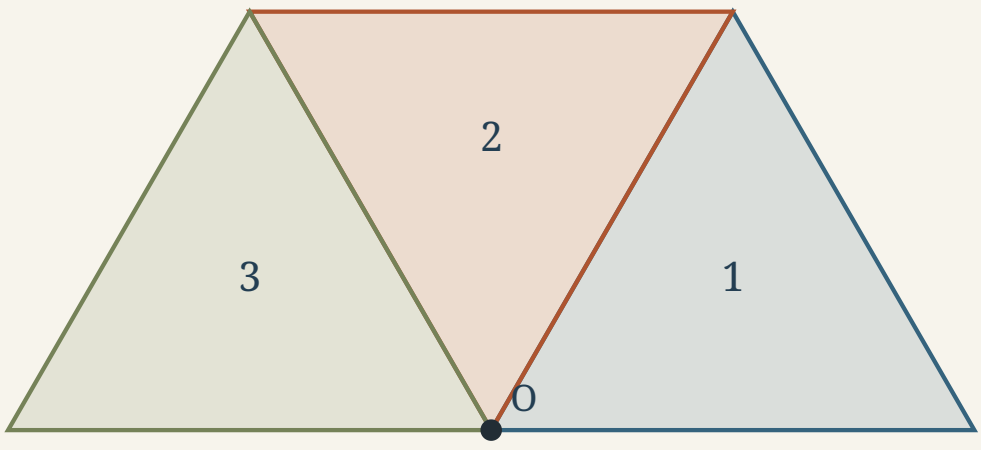
The addendum after Euclid XIII.18, Heiberg’s 1885 Greek edition and Latin translation, printed pp.336–341; XI definition 11 and XI.21, pp.4–5 and 54–59. Selected excerpts, new English and contextual notes. Exact angle arithmetic, coordinate models and the two-tetrahedra comparison are modern teaching work. The five-case classification assumes a closed convex polyhedron, congruent regular faces, and the same number of faces at every vertex. Equal regular faces alone do not impose that last condition.

The chosen corner

3 × 60° = 180° · deficit 180°

3 × 60° = 180°. A 180° deficit passes the local angle test. A complete uniform example is the tetrahedron.

Selected model: Triangular bipyramid; vertex 3; yaw 11°, pitch 6°. The angle arithmetic describes face angles at a vertex, not steradians or dihedral angles.



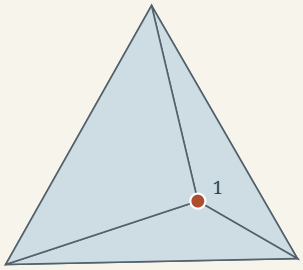
3 faces · 60° at O on each face

The drawing lays actual regular polygon faces in one plane. A gap records missing angle; closing it would move faces out of the plane. An overlap depicts excess angle. A positive deficit alone does not construct a complete solid or establish its uniqueness.

For q ≥ 3 and p ≥ 6, each face contributes at least 120°; the sum is at least 360°. Only p = 3, 4 or 5 can survive. Two faces are excluded by XI definition 11. The five surviving uniform pairs are Tetrahedron: p=3, q=3; Octahedron: p=3, q=4; Icosahedron: p=3, q=5; Cube: p=4, q=3; Dodecahedron: p=5, q=3.

Equal faces do not force equal corners

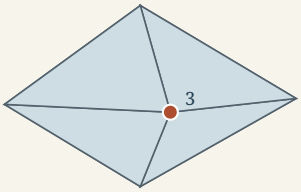
Tetrahedron



Tetrahedron · F 4 / E 6 / V 4

4 congruent regular faces. Every vertex has 3 incident faces.

Two joined tetrahedra



Triangular bipyramid · F 6 / E 9 / V 5

Six equal equilateral outside faces. Two tips have three incident faces; three belt vertices have four. The joined base pair is inside.

Join two regular tetrahedra along one congruent triangular base. Their two base faces are internal; the outside has six congruent equilateral triangles, nine edges and five vertices. Every edge has length 1 in the bipyramid coordinates. Two tip vertices have three faces and 180° deficits; three belt vertices have four faces and 120° deficits. The face condition alone therefore admits another convex solid. This modern comparison does not decide the ancient author's intended meaning or the textual history of the addendum.

Actual face incidences in the triangular bipyramid

Vertex	Faces at it	Face-angle sum	Deficit
1	3	180°	180°
2	3	180°	180°
3	4	240°	120°
4	4	240°	120°

Vertex	Faces at it	Face-angle sum	Deficit
5	4	240°	120°

Edge lengths: 1.000000–1.000000 coordinate units; F − E + V = 2. Rounded coordinates illustrate the construction; the angle classification uses exact integer arithmetic.

Can this one corner count the whole solid?

If every vertex had this same 120° deficit, the modern 720° total-deficit relation would give V = 720 / 120 = 6. Counting each edge at its two endpoints gives E = qV/2; counting each face at its p corners gives F = qV/p.

Conditional uniform counts and the actual triangular bipyramid

Count	If every vertex matched	Actual model
Vertices V	6	5
Edges E	12	9
Faces F	8	6

The uniformity condition fails in the bipyramid. Selecting a tip conditionally predicts four vertices; selecting a belt vertex predicts six. The actual solid has five. A local corner cannot supply the missing global condition.

Modern explanation, not part of the selected ancient passage: a closed convex polyhedron satisfies V − E + F = 2. Summing the face angles gives Σδ = 360V − 180(2E − 2F) = 720°. This model’s actual deficits total 720°. The total does not require all vertices to be alike.

With a common p and q at every vertex, the necessary face-angle condition leaves five candidate pairs. The complete displayed solids supply examples. Their coordinate meshes and the modern incidence checks are not reconstructions of the constructions in XIII.13–17 or a formal proof of global uniqueness.

Read the argument

Addendum after XIII.18 · opening sentence

1. Which figures does the sentence describe?

GREEK · P.336

Λέγω δή, ὅτι παρὰ τὰ εἰρημένα πέντε σχήματα οὐ συσταθήσεται ἕτερον σχῆμα περιεχόμενον ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίων ἴσων ἀλλήλοις.

HEIBERG'S LATIN · P.337

Iam dico, praeter quinque figuras, quas nominauimus, nullam aliam construi posse polygonis et aequilateris et aequiangulis inter se aequalibus comprehensam.

> I now say that, besides the five figures already mentioned, no other figure will be constructed that is enclosed by equilateral and equiangular figures equal to one another.

ἰσοπλεύρων and ἰσογωνίων describe the enclosing faces; ἴσων ἀλλήλοις says those faces are equal to each other. This sentence does not separately state that the same number of faces must meet at every vertex. Our joined tetrahedra test the face condition alone. This is an examination of the stated conditions, not a claim about the original author's intended definition or the date of the addendum.

ἴσων ἀλλήλοις / inter se aequalibus

equal to one another; genitive plural in the Greek, describing the enclosing figures.

Addendum · first reason

2. Two planes are not a closed corner

GREEK · P.336

Ὑπὸ μὲν γὰρ δύο τριγώνων ἢ ὅλως ἐπιπέδων στερεὰ γωνία οὐ συνίσταται.

HEIBERG'S LATIN · P.337

Nam ex duobus triangulis aut omnino figuris planis angulus solidus construi nequit.

> For a solid angle is not formed by two triangles, or by two plane figures in general.

A hinge made from two sheets is not yet the enclosed polyhedral corner meant here. XI definition 11 explicitly requires more than two plane angles, meeting at one point and not lying in one plane. Its complete Greek and Latin pages are retained. The app therefore refuses q = 2 even when the angle sum is less than a full turn.

στερεὰ γωνία / angulus solidus

a solid corner, made by plane face angles. Its angle sum here is not a solid-angle measurement in steradians.

Addendum · the triangular cases

3. Three, four, five

GREEK · P.336

ὑπὸ δὲ τριῶν τριγώνων ἡ τῆς πυραμίδος, ὑπὸ δὲ τεσσάρων ἡ τοῦ ὀκταέδρου, ὑπὸ δὲ πέντε ἡ τοῦ εἰκοσαέδρου·

HEIBERG'S LATIN · P.337

ex tribus uero triangulis angulus pyramidis construitur, ex quattuor octaedri, ex quinque icosaedri.

> From three triangles comes the corner of the pyramid; from four, that of the octahedron; from five, that of the icosahedron.

The repeated γωνία is understood after the first sentence: these are corners, not counts of all faces in a solid. The regular tetrahedron has four faces in total, with three at each corner. The six-faced bipyramid has both three-face and four-face corners.

τριῶν / τεσσάρων / πέντε / tribus / quattuor / quinque

three / four / five, counted at a single vertex. Do not confuse these with a polyhedron's total face count.

Addendum · selected clause

4. The sixth triangle uses the entire turn

GREEK · P.336

οὔσης γὰρ τῆς τοῦ ἰσοπλεύρου τριγώνου γωνίας διμοίρου ὀρθῆς ἔσονται αἱ ἓξ τέσσαρσιν ὀρθαῖς ἴσαι· ὅπερ ἀδύνατον·

HEIBERG'S LATIN · P.337 AND P.339

nam cum angulus trianguli aequilateri duae partes sint recti, sex anguli quattuor rectis aequales erunt; quod fieri non potest;

> For, since the angle of an equilateral triangle is two thirds of a right angle, the six angles will equal four right angles—which is impossible here.

διμοίρου ὀρθῆς means two thirds of a right angle, not two right angles. Six times 60° is 360°. The six triangles can lie together in a plane, but they do not make the strictly convex corner under discussion. The final Latin words continue at the top of p.339; the page selector can show that continuation.

διμοίρου ὀρθῆς / duae partes ... recti

two thirds of a right angle: 2/3 × 90° = 60°. Degrees and multiplication signs are our notation.

Addendum · appeal to XI.21

5. Less than four right angles

GREEK · P.338

ἅπασα γὰρ στερεὰ γωνία ὑπὸ ἐλασσόνων ἢ τεσσάρων ὀρθῶν περιέχεται.

HEIBERG'S LATIN · P.339

nam omnis angulus solidus minus quattuor rectis comprehenditur.

> For every solid angle is enclosed by face angles whose sum is less than four right angles.

This is a strict inequality for the convex corners studied here. A positive angular deficit is a necessary local condition, not a construction of a complete polyhedron. XI.21's printed proof treats a three-face corner; Heiberg's p.59 footnote says to reason similarly for more faces. We retain both the proof and that editorial note, rather than present the on-screen arithmetic as its ancient proof.

ἐλασσόνων ἢ / minus

less than; equality is excluded. The ἢ here introduces the comparison, not an alternative.

Addendum · square faces

6. A fourth square has no turn left

GREEK · P.338

ὑπὸ δὲ τετραγώνων τριῶν ἡ τοῦ κύβου γωνία περιέχεται· ὑπὸ δὲ τεσσάρων ἀδύνατον· ἔσονται γὰρ πάλιν τέσσαρες ὀρθαί.

HEIBERG'S LATIN · P.339

tribus autem quadratis angulus cubi comprehenditur. quattuor autem nullus; nam rursus quattuor recti erunt.

> The corner of the cube is enclosed by three squares. Four cannot enclose such a corner: once again there would be four right angles.

Each square contributes one right angle. Three give 270° and leave a 90° deficit; four give 360°. The deficit is the missing planar angle when the corner is cut open and laid flat, not the dihedral angle between two faces.

τετραγώνων / quadratis

squares, in the genitive plural after ὑπό. 'Four-angled' alone would not distinguish an arbitrary quadrilateral; the context supplies regularity.

Addendum · selected pentagon clause

7. The pentagon makes the excess visible

GREEK · P.338

οὔσης γὰρ τῆς τοῦ πενταγώνου ἰσοπλεύρου γωνίας ὀρθῆς καὶ πέμπτου, ἔσονται αἱ τέσσαρες γωνίαι τεσσάρων ὀρθῶν μείζους· ὅπερ ἀδύνατον.

HEIBERG'S LATIN · P.339

nam cum angulus pentagoni aequilateri aequalis sit recto angulo cum quinta parte recti, quattuor anguli quattuor rectis maiores erunt; quod fieri non potest.

> For, since the angle of the equilateral pentagon is a right angle and a fifth of a right angle, four such angles will exceed four right angles—which is impossible here.

The regular pentagon contributes 108°. Three leave 36°; four exceed a full turn by 72°. The selected clause abbreviates 'equilateral and equiangular' from the surrounding argument: this is not a claim about every equilateral pentagon. The following lemma proves the 108° result using five central triangles; its complete pages are retained.

ὀρθῆς καὶ πέμπτου / recto angulo cum quinta parte recti

a right angle plus one fifth of a right angle: 90° + 18° = 108°.

XI · definition 11, second formulation

8. A corner needs more than two face angles

GREEK · P.4

Ἄλλως· στερεὰ γωνία ἐστὶν ἡ ὑπὸ πλειόνων ἢ δύο γωνιῶν ἐπιπέδων περιεχομένη μὴ οὐσῶν ἐν τῷ αὐτῷ ἐπιπέδῳ πρὸς ἑνὶ σημείῳ συνισταμένων.

HEIBERG'S LATIN · P.5

Aliter. Solidus angulus est, qui amplius quam duobus angulis planis continetur non in eodem plano positis et ad unum punctum coniunctis.

> Alternatively: a solid angle is enclosed by more than two plane angles, not lying in the same plane, coming together at one point.

This is the definition's second formulation, introduced by Ἄλλως / Aliter. The complete page also retains the first formulation and editorial apparatus. Read ‘plane angles’ as the corners of the faces; ‘not in the same plane’ excludes the flat six-triangle arrangement.

πρὸς ἑνὶ σημείῳ / ad unum punctum

at one point; the faces are counted at the shared vertex, not scattered over the whole object.

Lemma following the addendum · selected statement

9. Equilateral and equiangular both matter

GREEK · P.338

Ὅτι δὲ ἡ τοῦ ἰσοπλεύρου καὶ ἰσογωνίου πενταγώνου γωνία ὀρθή ἐστι καὶ πέμπτου, οὕτω δεικτέον.

HEIBERG'S LATIN · P.339

Angulum autem pentagoni aequilateri et aequianguli aequalem esse angulo recto et quintae parti recti, sic demonstrandum.

> That the angle of an equilateral and equiangular pentagon is a right angle and a fifth must be shown as follows.

A pentagon with equal sides alone can have unequal angles. The two adjectives supply both conditions. Joining the centre to five vertices gives central angles of 72°, with two equal base angles of 54° in each triangle; two neighbouring base angles give the face angle of 108°. This calculation explains the retained lemma; the modern 72°, 54° and 108° labels are not printed in its Greek text.

ἰσοπλεύρου καὶ ἰσογωνίου / aequilateri et aequianguli

equal-sided and equal-angled; both are required for a regular face.

Retained files and a paper experiment

The CSV enumerates the 70 supported p/q choices; it does not assert that the learner tried them. A separate section retains the selected mesh's actual vertex incidences. The A4 cutting sheet has two four-face nets and a 50 mm calibration bar. Tape their free edges, then place the marked bases together. Physical printing and assembly are unverified.


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  }
}
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Reopen these settings in the corresponding workshop. This link requires the app; this reading's diagrams, pages, font and native downloads need no scripts or network subresources.

The complete printed evidence

J. L. Heiberg, *Euclidis Elementa*, volume IV, Leipzig: B. G. Teubner, 1885. Title, XI definition 11 on pp.4–5, XI.21 on pp.54–59, and the addendum and following lemma on pp.336–341. Digitized Internet Archive copy. The title is PDF page 7; numbered pages are PDF page number minus 10. Whole unretouched pages are reduced to a 2200 px long edge. Text, editorial notes and apparatus remain together.

Selected passages remove line-end hyphenation, normalize printed theta glyphs and join line breaks. Greek/Latin punctuation is lightly regularized; Latin u/v spellings remain. Superscript reference markers are omitted from the selections. The printed Latin clause in reading 4 continues onto p.339. The addendum's opening is after a divider on pp.336–337; it is not the opening proposition of XIII.18. The lemma ends on pp.340–341. New English and language notes have not had independent philological review.

Original public-domain volume. The addendum after Euclid XIII.18, Heiberg’s 1885 Greek edition and Latin translation, printed pp.336–341; XI definition 11 and XI.21, pp.4–5 and 54–59. Selected excerpts, new English and contextual notes. Exact angle arithmetic, coordinate models and the two-tetrahedra comparison are modern teaching work. The five-case classification assumes a closed convex polyhedron, congruent regular faces, and the same number of faces at every vertex. Equal regular faces alone do not impose that last condition.

EUCLIDIS

ELEMENTA.

EDIDIT ET LATINE INTERPRETATUS EST

I. L. HEIBERG,

DR. PHIL.

UOL. IV.

LIBROS XI—XIII CONTINENS.

LIPSIAE

IN AEDIBUS B. G. TEUBNERI.

MDCCCLXXXV.

Title · PDF p.7 · SHA-256 f4afb75d2d0ddf260fac76689d86285d9ed19304ecab5fbbe11f8edca2539d27

ζ΄. Ἐπίπεδον πρὸς ἐπίπεδον ὁμοίως κεκλίσθαι λέγεται καὶ ἕτερον πρὸς ἕτερον, ὅταν αἱ εἰρημέναι τῶν κλίσεων γωνίαι ἴσαι ἀλλήλαις ὦσιν.

η΄. Παράλληλα ἐπίπεδά ἐστι τὰ ἀσύμπτωτα.

θ΄. Ὅμοια στερεὰ σχήματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεχόμενα ἴσων τὸ πλῆθος.

ι΄. Ἴσα δὲ καὶ ὅμοια στερεὰ σχήματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεχόμενα ἴσων τῷ πλήθει καὶ τῷ μεγέθει.

ια΄. Στερεὰ γωνία ἐστὶν ἡ ὑπὸ πλειόνων ἢ δύο γραμμῶν ἁπτομένων ἀλλήλων καὶ μὴ ἐν τῇ αὐτῇ ἐπιφανείᾳ οὐσῶν πρὸς πάσαις ταῖς γραμμαῖς κλίσις. Ἄλλως· στερεὰ γωνία ἐστὶν ἡ ὑπὸ πλειόνων ἢ δύο γωνιῶν ἐπιπέδων περιεχομένη μὴ οὐσῶν ἐν τῷ αὐτῷ ἐπιπέδῳ πρὸς ἑνὶ σημείῳ συνισταμένων.

ιβ΄. Πυραμίς ἐστι σχῆμα στερεὸν ἐπιπέδοις περιχόμενον ἀπὸ ἑνὸς ἐπιπέδου πρὸς ἑνὶ σημείῳ συνεστώς.

ιγ΄. Πρίσμα ἐστὶ σχῆμα στερεὸν ἐπιπέδοις περιεχόμενον, ὧν δύο τὰ ἀπεναντίον ἴσα τε καὶ ὅμοιά ἐστι καὶ παράλληλα, τὰ δὲ λοιπὰ παραλληλόγραμμα.

ιδ΄. Σφαῖρά ἐστιν, ὅταν ἡμικυκλίου μενούσης τῆς διαμέτρου περιενεχθὲν τὸ ἡμικύκλιον εἰς τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρξατο φέρεσθαι, τὸ περιληφθὲν σχῆμα.

ιε΄. Ἄξων δὲ τῆς σφαίρας ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ ἡμικύκλιον στρέφεται.

ις΄. Κέντρον δὲ τῆς σφαίρας ἐστὶ τὸ αὐτό, ὅ καὶ τοῦ ἡμικυκλίου.

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8. Hero def. 115, 2. 9. ib. 118, 2. 11. ib. 24. 12. ib. 100. 14. ib. 77. 11—15. Psellus p. 49—50.

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3. ωσι Vb. 4. παράλληλα ἐπί- in ras., -πεδα mg. m. 2 V. 5. ὑπό] corr. ex ἀπό m. 1 b. 12. πρός] B; ἡ πρός

Printed p.4 · PDF p.14 · SHA-256 4d0210229e42d4571e25aa820aa8e776d3a8a3689df66663a514489aa206ab40

7. Planum ad planum similiter inclinatum dicitur atque aliud planum ad aliud, ubi anguli inclinationum, quos definiuimus, aequales sunt inter se.

8. Parallela plana sunt, quae non concurrunt.

9. Similes figurae solidae sunt, quae planis similibus continentur numero aequalibus.

10. Aequales autem et similes figurae solidae sunt, quae planis similibus continentur et numero et magnitudine aequalibus.

11. Solidus angulus est amplius quam duarum rectarum inter se tangentium nec in eadem superficie positarum ad omnes rectas inclinatio.[1]) Aliter. Solidus angulus est, qui amplius quam duobus angulis planis continetur non in eodem plano positis et ad unum punctum coniunctis.

12. Pyramis est figura solida planis comprehensa, quae ab uno plano ad unum punctum componitur.

13. Prisma est figura solida planis comprehensa, quorum duo opposita et aequalia et similia sunt, reliqua autem parallelogramma.

14. Sphaera est figura comprehensa, ubi manente diametro semicirculi semicirculus circumactus rursus ad eundem locum restituitur, unde ferri coeptus est.

15. Axis autem sphaerae est recta manens, circum quam semicirculus circumagitur.

16. Centrum autem sphaerae idem est ac semicirculi.

1) Haec definitio, quae loquendi genere ab Euclide abhorret, fortasse ex Elementis antiquioribus ab eo desumpta est.

PFVb. 13. *ἐπιπέδων'' γωνιῶν'* F, *ἐπιπέδων γωνιῶν* B. 15. Ante *ἑνί* del. *εν* F. 17. *συνεστός* Bb; in P non liquet. 18. *ἐστίν* PF. 19. *ὧν*] om. *φ*. 20. *ἐστιν* F. 22. *τὸ ἡμικύκλιον*] mg. m, 1 b. 26. *ἐστίν* F.

Printed p.5 · PDF p.15 · SHA-256 f3ee32bdea92c1acb26a70c5423219e65ab0cadcee3d2506f95990ca8a8f699c

διὰ τῶν $BAΓ$ ἐπιπέδῳ ἴση ἡ ὑπὸ BAE, καὶ κείσθω τῇ $AΔ$ ἴση ἡ AE, καὶ διὰ τοῦ E σημείου διαχθεῖσα ἡ $BEΓ$ τεμνέτω τὰς AB, $AΓ$ εὐθείας κατὰ τὰ B, $Γ$ σημεῖα, καὶ ἐπεζεύχθωσαν αἱ $ΔB$, $ΔΓ$. καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΔA$ τῇ AE, κοινὴ δὲ ἡ AB, δύο δυσὶν ἴσαι· καὶ γωνία ἡ ὑπὸ $ΔAB$ γωνίᾳ τῇ ὑπὸ BAE ἴση· βάσις ἄρα ἡ $ΔB$ βάσει τῇ BE ἐστιν ἴση. καὶ ἐπεὶ δύο αἱ $BΔ$, $ΔΓ$ τῆς $BΓ$ μείζονές εἰσιν, ὧν ἡ $ΔB$ τῇ BE ἐδείχθη ἴση, λοιπὴ ἄρα ἡ $ΔΓ$ λοιπῆς τῆς $EΓ$ μείζων ἐστίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΔA$ τῇ AE, κοινὴ δὲ ἡ $AΓ$, καὶ βάσις ἡ $ΔΓ$ βάσεως τῆς $EΓ$ μείζων ἐστίν, γωνία ἄρα ἡ ὑπὸ $ΔAΓ$ γωνίας τῆς ὑπὸ $EAΓ$ μείζων ἐστίν. ἐδείχθη δὲ καὶ ἡ ὑπὸ $ΔAB$ τῇ ὑπὸ BAE ἴση· αἱ ἄρα ὑπὸ $ΔAB$, $ΔAΓ$ τῆς ὑπὸ $BAΓ$ μείζονές εἰσιν. ὁμοίως δὴ δείξομεν, ὅτι καὶ αἱ λοιπαὶ σύνδυο λαμβανόμεναι τῆς λοιπῆς μείζονές εἰσιν.

Ἐὰν ἄρα στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέχηται, δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.

κα'.

Ἅπασα στερεὰ γωνία ὑπὸ ἐλασσόνων [ἢ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέχεται.

Ἔστω στερεὰ γωνία ἡ πρὸς τῷ A περιεχομένη ὑπὸ ἐπιπέδων γωνιῶν τῶν ὑπὸ $BAΓ$, $ΓAΔ$, $ΔAB$· λέγω, ὅτι αἱ ὑπὸ $BAΓ$, $ΓAΔ$, $ΔAB$ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

---

1. ἐπιπέδῳ] in ras. m. 1 P. ἡ] supra scr. V, ut lin. 2. κείσθω τῇ] διὰ τοῦ E ση φ (non F). Hinc plerasque ineptias manus φ omisi, maxime ubi aut certa uestigia ueri supererant, aut certe nulla erat causa de scriptura cod. F dubitandi.

Printed p.54 · PDF p.64 · SHA-256 f452ecf08dca4bb42295b334a7402c60d9d413aad60a01afe937e703907c045a

aequalis construatur $\angle BAE$, et ponatur $AE = A\Delta$, et $BE\Gamma$ per punctum E ducta rectas AB, $A\Gamma$ secet in B, Γ punctis, et ducantur ΔB, $\Delta\Gamma$. et quoniam $\Delta A = AE$, et AB communis est, duo latera duobus aequalia sunt; et $\angle \Delta AB = BAE$. itaque $\Delta B = BE$ [I, 4]. et quoniam $B\Delta + \Delta\Gamma > B\Gamma$ [I, 20], et demonstratum est, esse $\Delta B = BE$, erit $\Delta\Gamma > E\Gamma$. et quoniam $\Delta A = AE$, et $A\Gamma$ communis est, et $\Delta\Gamma > E\Gamma$, erit $\angle \Delta A\Gamma > EA\Gamma$ [I, 25]. et demonstratum est, esse etiam $\angle \Delta AB = BAE$. itaque $\Delta AB + \Delta A\Gamma > BA\Gamma$. eodem modo demonstrabimus, etiam reliquos angulos duo simul coniunctos reliquo maiores esse.

Ergo si angulus solidus tribus angulis planis continetur, duo quilibet reliquo maiores sunt quoquo modo coniuncti; quod erat demonstrandum.

XXI.

Omnis angulus solidus planis angulis minoribus, quam sunt quattuor anguli recti, continentur.

Sit angulus solidus, qui ad A positus est, comprehensus planis angulis $BA\Gamma$, $\Gamma A\Delta$, ΔAB. dico, esse $BA\Gamma + \Gamma A\Delta + \Delta AB$ minores quattuor rectis.

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3. Γ] corr. ex E m. 1 b. 4. ΔB] $B\Delta$ F. 6. Post ἴσαι ras. 4 litt. hab. V. 7. ἐστιν ἴση] ἴση seq. spatio uacuo 3 litt. V. 8. $B\Delta$] $B''\Delta'$ b, ΔB BV. τῇ V? 10. ἐστίν] (prius) ἐστί PBV, comp. Fb. AE] in ras. V. 11. $\Delta\Gamma$] corr. ex ΔE B. 12. ἐστί PBV, comp. F. Dein add. καί V. $\Delta A\Gamma$] $\Delta B\Gamma$ φ. 14. τῆς] bis P, corr. m. 1; τοῖς F. 17. ὑπό — 19. δεῖξαι] καὶ τὰ ἑξῆς V. 21. ὑπό] corr. ex ἀπό P. ἤ] om. P. 22. ἐπιπέδων ὀρθῶν γωνιῶν V. 23. τῷ] corr. in τό m. 1 b. 24. ὑπό — 25. αἱ] mg. m. 2 B. 24. $\Gamma A\Delta$] $\Delta A\Gamma$ φ et in ras. V. 25. ὑπό] eras. B; m. 2 V. $\Gamma A\Delta$] F m. 1, $\Delta A\Gamma$ F m. 2 et V in ras. 26. εἰσι V.

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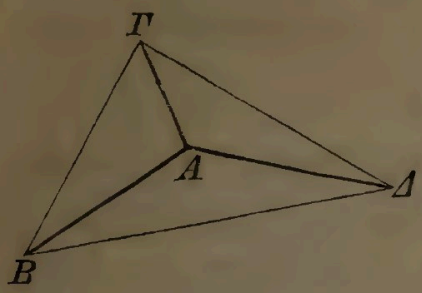
Εἰλήφθω γὰρ ἐφ' ἑκάστης τῶν ΑΒ, ΑΓ, ΑΔ τυχόντα σημεῖα τὰ Β, Γ, Δ, καὶ ἐπεζεύχθωσαν αἱ ΒΓ, ΓΔ, ΔΒ. καὶ ἐπεὶ στερεὰ γωνία ἡ πρὸς τῷ Β ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέχεται τῶν ὑπὸ ΓΒΑ, ΑΒΔ, ΓΒΔ, δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσιν· αἱ ἄρα ὑπὸ ΓΒΑ, ΑΒΔ τῆς ὑπὸ ΓΒΔ μείζονές εἰσιν. διὰ τὰ αὐτὰ δὴ καὶ αἱ μὲν ὑπὸ ΒΓΑ, ΑΓΔ τῆς ὑπὸ ΒΓΔ μείζονές εἰσιν, αἱ δὲ ὑπὸ ΓΔΑ, ΑΔΒ τῆς ὑπὸ ΓΔΒ μείζονές εἰσιν· αἱ ἓξ ἄρα γωνίαι αἱ ὑπὸ ΓΒΑ, ΑΒΔ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ τριῶν τῶν ὑπὸ ΓΒΔ, ΒΓΔ, ΓΔΒ μείζονές εἰσιν. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ ΓΒΔ, ΒΔΓ, ΒΓΔ δυσὶν ὀρθαῖς ἴσαι εἰσίν· αἱ ἓξ ἄρα αἱ υπὸ ΓΒΑ, ΑΒΔ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ δύο ὀρθῶν μείζονές εἰσιν. καὶ ἐπεὶ ἑκάστου τῶν ΑΒΓ, ΑΓΔ, ΑΔΒ τριγώνων αἱ τρεῖς γωνίαι δυσὶν ὀρθαῖς ἴσαι εἰσίν, αἱ ἄρα τῶν τριῶν τριγώνων ἐννέα γωνίαι αἱ ὑπὸ ΓΒΑ, ΑΓΒ, ΒΑΓ, ΑΓΔ, ΓΔΑ, ΓΑΔ, ΑΔΒ, ΔΒΑ, ΒΑΔ ἓξ ὀρθαῖς ἴσαι εἰσίν, ὧν αἱ ὑπὸ ΑΒΓ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ, ΔΒΑ ἓξ γωνίαι δύο ὀρθῶν εἰσι μείζονες· λοιπαὶ ἄρα αἱ ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ τρεῖς [γωνίαι] περιέχουσαι τὴν στερεὰν γωνίαν τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

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2. *Γ*] supra scr. m. 1 V. 3. *ΔΒ*] *ΑΒ* φ. 4. Ante *τριῶν* ins. *γάρ* m. 2 V. 5. *ΓΒΔ*] in ras. m. 1 P. 6. *ὑπό*] (alt.) om. F. *εἰσι* BV, comp. Fb. 7. *ΒΓΑ*] supra *Α* scr. *Δ* m. 1 b. 8. *ΒΓΔ*] *ΓΒΔ* F, corr. m. 2 (sed euan.). *εἰσι* BVb, comp. F. *αἱ δέ*] *καὶ ἔτι αἱ* BFVb. 10. *ΑΒΔ*] *ΒΔ* in ras. B, item litt. seq. *ΓΔΑ*] in ras. V. 11. *ΒΓΔ*] *ΓΔ* in ras. V. *ΓΔΒ*] in ras. V. *ἀλλ'* b. 12. *ΒΓΔ*] *Β* et *Δ* in ras. V. *εἰσί* V, comp. F. 13. *ΑΒΔ*] m. rec. V. *ΓΔΑ*] in ras. V; *ΑΔΓ* e corr. m. 2 B. 14. *δύο*] *ΑΒΔ δύο* V. *εἰσι* BVb, comp. F. 15. *αἱ τρεῖς τριγώνων* F, corr. m. 1. *τριγώνου* P, et b, sed corr. m. 1. 17. *ΓΒΑ*] *ΓΒΔ* F, *ΒΑ* e corr. V. *ΑΓΒ*] *ΑΒΓ* P. 18. *ΓΔΑ*]

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sumatur enim in singulis rectis AB, $A\Gamma$, $A\Delta$ quaelibet puncta B, Γ, Δ, et ducantur $B\Gamma$, $\Gamma\Delta$, ΔB.



et quoniam angulus solidus, qui ad B positus est, tribus angulis planis continetur ΓBA, $AB\Delta$, $\Gamma B\Delta$, duo quilibet reliquo maiores sunt [prop. XX]. itaque $\Gamma BA + AB\Delta > \Gamma B\Delta$. eadem de causa erunt etiam $B\Gamma A + A\Gamma\Delta > B\Gamma\Delta$, $\Gamma\Delta A + A\Delta B > \Gamma\Delta B$.

itaque $\Gamma BA + AB\Delta + B\Gamma A + A\Gamma\Delta + \Gamma\Delta A + A\Delta B > \Gamma B\Delta + B\Gamma\Delta + \Gamma\Delta B$. uerum

$$\Gamma B\Delta + B\Delta\Gamma + B\Gamma\Delta$$

duobus rectis aequales sunt [I, 32]. itaque sex anguli

$$\Gamma BA + AB\Delta + B\Gamma A + A\Gamma\Delta + \Gamma\Delta A + A\Delta B$$

duobus rectis maiores sunt. et quoniam singulorum triangulorum $AB\Gamma$, $A\Gamma\Delta$, $A\Delta B$ tres anguli duobus rectis aequales sunt, nouem anguli trium triangulorum $\Gamma BA + A\Gamma B + BA\Gamma + A\Gamma\Delta + \Gamma\Delta A + \Gamma A\Delta + A\Delta B + \Delta BA + BA\Delta$ sex rectis aequales sunt, quorum

$$AB\Gamma + B\Gamma A + A\Gamma\Delta + \Gamma\Delta A + A\Delta B + \Delta BA$$

duobus rectis maiores sunt. itaque reliqui

$$BA\Gamma + \Gamma A\Delta + \Delta AB,$$

qui angulum solidum continent, quattuor rectis minores sunt.

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in ras. V; $\Delta A\Gamma$ B. $\Gamma A\Delta$] $A\Delta$ in ras. V; $\Gamma\Delta A$ B. $BA\Delta$] $B\Delta A$ P. 20. *μείζονές εἰσι(ν)* BV. 21. *γωνίαι*] om. P. 22. *εἰσι* V, comp. F. Seq. in V *πάντη*, sed del.

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Ἅπασα ἄρα στερεὰ γωνία ὑπὸ ἐλασσόνων [ἢ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέχεται· ὅπερ ἔδει δεῖξαι.

κβ'.

Ἐὰν ὦσι τρεῖς γωνίαι ἐπίπεδοι, ὧν αἱ δύο τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, περιέχωσι δὲ αὐτὰς ἴσαι εὐθεῖαι, δυνατόν ἐστιν ἐκ τῶν ἐπιζευγνυουσῶν τὰς ἴσας εὐθείας τρίγωνον συστήσασθαι.

Ἔστωσαν τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, ὧν αἱ δύο τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν ὑπὸ ΑΒΓ, ΔΕΖ τῆς ὑπὸ ΗΘΚ, αἱ δὲ ὑπὸ ΔΕΖ, ΗΘΚ τῆς ὑπὸ ΑΒΓ, καὶ ἔτι αἱ ὑπὸ ΗΘΚ, ΑΒΓ τῆς ὑπὸ ΔΕΖ, καὶ ἔστωσαν ἴσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ εὐθεῖαι, καὶ ἐπεζεύχθωσαν αἱ ΑΓ, ΔΖ, ΗΚ· λέγω, ὅτι δυνατόν ἐστιν ἐκ τῶν ἴσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι, τουτέστιν ὅτι τῶν ΑΓ, ΔΖ, ΗΚ δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσιν.

Εἰ μὲν οὖν αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ γωνίαι ἴσαι ἀλλήλαις εἰσίν, φανερόν, ὅτι καὶ τῶν ΑΓ, ΔΖ, ΗΚ ἴσων γινομένων δυνατόν ἐστιν ἐκ τῶν ἴσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. εἰ δὲ οὔ,

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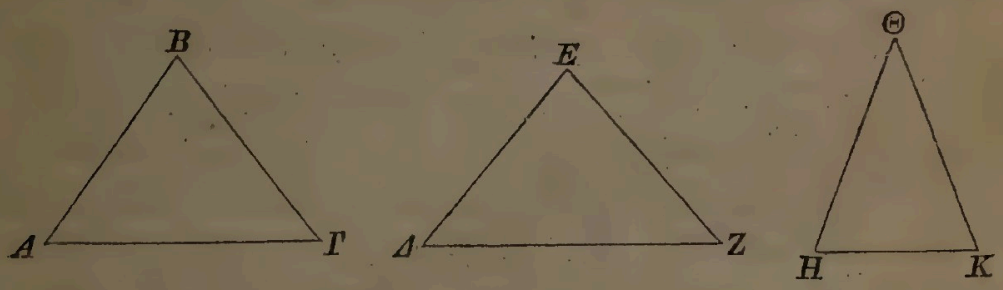
1. ἄρα] supra scr. m. 1 P. ὑπό — 3. δεῖξαι]: ~ V. 1. ἢ] postea add. m. 1 P. 7. περιέχωσιν P, περιέχουσι F. 8. Supra ἴσας add. γωνίας m. 2 B, del. m. rec. εὐθείας] γωνίας εὐθειῶν V. 11. εἰσι] ἔστωσαν BFV et b (εσ- in ras.). 15. εὐθεῖαι] m. rec. V. 17. συνστήσασθαι P, corr. m. 2. 18. ὅτι] corr. ex τό m. 2 F. 19. μείζους V. εἰσι πάντῃ μεταλαμβανόμεναι Theon (BFVb). 21. εἰσι ἴσαι V. εἰσίν] εἰσί PBb, comp. F.; om. V. 22. γιγνομένων F, γενομένων b.

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Ergo omnis[1]) angulus solidus planis angulis minoribus, quam sunt quattuor recti, continetur; quod erat demonstrandum.

XXII.

Si tres anguli plani sunt, quorum duo reliquo maiores sunt quoquo modo coniuncti, et eos aequales



continent rectae, fieri potest, ut ex rectis aequales rectas coniungentibus triangulus construatur.

Sint tres anguli plani $AB\Gamma$, ΔEZ, $H\Theta K$, quorum duo reliquo maiores sunt quoquo modo coniuncti,

$$AB\Gamma + \Delta EZ > H\Theta K,\ \Delta EZ + H\Theta K > AB\Gamma,$$
$$H\Theta K + AB\Gamma > \Delta EZ,$$

et sit $AB = B\Gamma = \Delta E = EZ = H\Theta = \Theta K$, et ducantur $A\Gamma$, ΔZ, HK. dico, fieri posse, ut ex rectis aequalibus rectis $A\Gamma$, ΔZ, HK triangulus construatur, hoc est, rectarum $A\Gamma$, ΔZ, HK duas quaslibet reliqua maiores esse.

iam si anguli $AB\Gamma$, ΔEZ, $H\Theta K$ inter se aequales sunt, manifestum est, cum etiam $A\Gamma$, ΔZ, HK aequales sint [I, 4], fieri posse, ut ex rectis aequalibus rectis $A\Gamma$, ΔZ, HK triangulus construatur. sin minus, in-

1) Nam in angulis solidis, qui plus quam tribus planis angulis continentur, similiter ratiocinandum est.

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τῆς $B\Delta$. τριπλῆ δὲ ἡ AB τῆς $B\Delta$· τριπλάσιον ἄρα τὸ ἀπὸ τῆς ZB τοῦ ἀπὸ τῆς $B\Delta$. ἔστι δὲ καὶ τὸ ἀπὸ τῆς $A\Delta$ τοῦ ἀπὸ τῆς ΔB τετραπλάσιον· διπλῆ γὰρ ἡ $A\Delta$ τῆς ΔB· μεῖζον ἄρα τὸ ἀπὸ τῆς $A\Delta$ τοῦ ἀπὸ τῆς ZB· μείζων ἄρα ἡ $A\Delta$ τῆς ZB· πολλῷ ἄρα ἡ $A\Lambda$ τῆς ZB μείζων ἐστίν. καὶ τῆς μὲν $A\Lambda$ ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ $K\Lambda$, ἐπειδήπερ ἡ μὲν ΛK ἐξαγώνου ἐστίν, ἡ δὲ $K\Lambda$ δεκαγώνου· τῆς δὲ ZB ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν η NB· μείζων ἄρα ἡ $K\Lambda$ τῆς NB. ἴση δὲ ἡ $K\Lambda$ τῇ ΛM· μείζων ἄρα ἡ ΛM τῆς NB [τῆς δὲ ΛM μείζων ἐστὶν ἡ MB]. πολλῷ ἄρα ἡ MB πλευρὰ οὖσα τοῦ εἰκοσαέδρου μείζων ἐστὶ τῆς NB πλευρᾶς οὔσης τοῦ δωδεκαέδρου· ὅπερ ἔδει δεῖξαι.

Λέγω δή, ὅτι παρὰ τὰ εἰρημένα πέντε σχήματα οὐ συσταθήσεται ἕτερον σχῆμα περιεχόμενον ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίων ἴσων ἀλλήλοις.

Ὑπὸ μὲν γὰρ δύο τριγώνων ἢ ὅλως ἐπιπέδων στερεὰ γωνία οὐ συνίσταται. ὑπὸ δὲ τριῶν τριγώνων ἡ τῆς πυραμίδος, ὑπὸ δὲ τεσσάρων ἡ τοῦ ὀκταέδρου, ὑπὸ δὲ πέντε ἡ τοῦ εἰκοσαέδρου· ὑπὸ δὲ ἓξ τριγώνων ἰσοπλεύρων τε καὶ ἰσογωνίων πρὸς ἑνὶ σημείῳ συνισταμένων οὐκ ἔσται στερεὰ γωνία· οὔσης γὰρ τῆς τοῦ ἰσοπλεύρου τριγώνου γωνίας διμοίρου ὀρθῆς ἔσονται αἱ ἓξ τέσσαρσιν ὀρθαῖς ἴσαι· ὅπερ ἀδύνατον·

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2. ἔστιν PB. 5. καὶ μείζων B. ἄρα καί V. τῆς ZB] (alt.) om. P. 6. ἐστί Vq. 7. τετμημένης V. 11. ΛM τῆς NB] in ras. m. 1 P. 12. τῆς δέ — MB] postea add. in mg. m. 1 P. 13. μείζω, ν add. m. 2 V. 14. Se-

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trario igitur $AB : B\Delta = ZB^2 : B\Delta^2$. uerum $AB = 3B\Delta$. itaque etiam $ZB^2 = 3B\Delta^2$. uerum etiam $A\Delta^2 = 4\Delta B^2$; nam $A\Delta = 2\Delta B$. itaque $A\Delta^2 > ZB^2$. quare $A\Delta > ZB$. itaque multo magis $A\Lambda > ZB$. et rectae $A\Lambda$ secundum rationem extremam ac mediam diuisae maior pars est $K\Lambda$, quoniam ΛK hexagoni est, KA autem decagoni [prop. IX]; rectae autem ZB secundum rationem extremam ac mediam diuisae maior pars est NB. itaque $K\Lambda > NB$. est autem $K\Lambda = \Lambda M$. quare $\Lambda M > NB$. ergo multo magis MB latus icosaedri NB latere dodecaedri maius est; quod erat demonstrandum.

Iam dico, praeter quinque figuras, quas nominauimus, nullam aliam construi posse polygonis et aequilateris et aequiangulis inter se aequalibus comprehensam.

Nam ex duobus triangulis aut omnino figuris planis angulus solidus construi nequit [XI def. 11]. ex tribus uero triangulis angulus pyramidis construitur, ex quattuor octaedri, ex quinque icosaedri. ex sex autem triangulis aequilateris et aequiangulis ad idem punctum coniunctis angulus solidus non orietur; nam cum angulus trianguli aequilateri duae partes sint recti, sex anguli quattuor rectis aequales erunt; quod fieri non

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Cum epimetro lin. 15 sq. cfr. Psellus p. 51 sq.

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quitur alia demonstratio extremae partis, u. app. 16. *συνσταθήσεται* P. 19. *ἢ ὅλως*] scripsi; ras. 2 uel 3 litt. P, supra scr. *ἀλλ' οὐδὲ ὑπὸ δύο* m. rec.; *ἀλλ' οὐδὲ ἄλλων δύο* Theon (BVq). 20. *οὐ*] om. Pq. 26. *αἱ*] om. q.

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ἅπασα γὰρ στερεὰ γωνία ὑπὸ ἐλασσόνων ἢ τεσσάρων ὀρθῶν περιέχεται. διὰ τὰ αὐτὰ δὴ οὐδὲ ὑπὸ πλειόνων ἢ ἐξ γωνιῶν ἐπιπέδων στερεὰ γωνία συνίσταται. ὑπὸ δὲ τετραγώνων τριῶν ἡ τοῦ κύβου γωνία περιέχεται· ὑπὸ δὲ τεσσάρων ἀδύνατον· ἔσονται γὰρ πάλιν τέσσαρες ὀρθαί. ὑπὸ δὲ πενταγώνων ἰσοπλεύρων καὶ ἰσογωνίων, ὑπὸ μὲν τριῶν ἡ τοῦ δωδεκαέδρου· ὑπὸ δὲ τεσσάρων ἀδύνατον· οὔσης γὰρ τῆς τοῦ πενταγώνου ἰσοπλεύρου γωνίας ὀρθῆς καὶ πέμπτου, ἔσονται αἱ τέσσαρες γωνίαι τεσσάρων ὀρθῶν μείζους· ὅπερ ἀδύνατον. οὐδὲ μὴν ὑπὸ πολυγώνων ἑτέρων σχημάτων περισχεθήσεται στερεὰ γωνία διὰ τὸ αὐτὸ ἄτοπον.

Οὐκ ἄρα παρὰ τὰ εἰρημένα πέντε σχήματα ἕτερον σχῆμα στερεὸν συσταθήσεται ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίων περιεχόμενον· ὅπερ ἔδει δεῖξαι.

Λῆμμα.

Ὅτι δὲ ἡ τοῦ ἰσοπλεύρου καὶ ἰσογωνίου πενταγώνου γωνία ὀρθή ἐστι καὶ πέμπτου, οὕτω δεικτέον.

Ἔστω γὰρ πεντάγωνον ἰσόπλευρον καὶ ἰσογώνιον τὸ ΑΒΓΔΕ, καὶ περιγεγράφθω περὶ αὐτὸ κύκλος ὁ ΑΒΓΔΕ, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον τὸ Ζ, καὶ ἐπεζεύχθωσαν αἱ ΖΑ, ΖΒ, ΖΓ, ΖΔ, ΖΕ. δίχα ἄρα τέμνουσι τὰς πρὸς τοῖς Α, Β, Γ, Δ, Ε τοῦ πεντα-

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2. ὀρθῶν γωνιῶν q. οὐδέ] om. q, οὐδ' P. 3. ἤ] om. P, supra scr. m. 1 B. γωνιῶν] τριγώνων q. 5. τέσσαρεις P. 8. δέ] om. q. ἰσοπλεύρου πενταγώνου V. 9. αἱ] supra m. rec. P. 10. τέσσαρες] -ες in ras. m. 1 P. In mg. m. 1 pro scholio: ὡς δείξει ὑποκάτω P. 11. πολυγωνίων π (non P). ἑτέρων] στερεῶν q. 12. αὐτό] om. BV.

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potest; nam omnis angulus solidus minus quattuor rectis comprehenditur [XI, 21]. eadem de causa ne ex pluribus quidem quam sex angulis planis solidus angulus construitur. tribus autem quadratis angulus cubi comprehenditur. quattuor autem nullus; nam rursus quattuor recti erunt. pentagonis autem aequilateris et aequiangulis tribus angulus dodecaedri comprehenditur, quattuor autem nullus; nam cum angulus pentagoni aequilateri aequalis sit recto angulo cum quinta parte recti, quattuor anguli quattuor rectis maiores erunt; quod fieri non potest. eadem de causa ne aliis quidem figuris polygonis angulus solidus comprehendetur.

ergo praeter quinque figuras, quas nominauimus, nulla alia figura solida construetur figuris aequilateris et aequiangulis comprehensa; quod erat demonstrandum.

Lemma.

Angulum autem pentagoni aequilateri et aequianguli aequalem esse angulo recto et quintae parti recti, sic demonstrandum.

sit enim pentagonum aequilaterum et aequiangulum $ABΓΔE$, et circum id circulus circumscribatur $ABΓΔE$ [IV, 14], et sumatur centrum eius Z [III, 1], et ducantur ZA, ZB, $ZΓ$, $ZΔ$, ZE. itaque angulos pentagoni ad A, B, $Γ$, $Δ$, E positos in binas partes aequales secant [I, 4]. et quoniam quinque anguli ad

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14. *συνσταθήσεται* P, corr. m. rec. 16. *λῆμμα*] om. codd.
17. *ὅτε* q. *τε καί* V. Post *ἰσογωνίου* add. *καί* q.
18. *ἐστιν* PB. *πέμπτον* q. 20. *τε καί* V. 22. *τό*] (prius) om. q. 24. *τέμνουσιν* PB.

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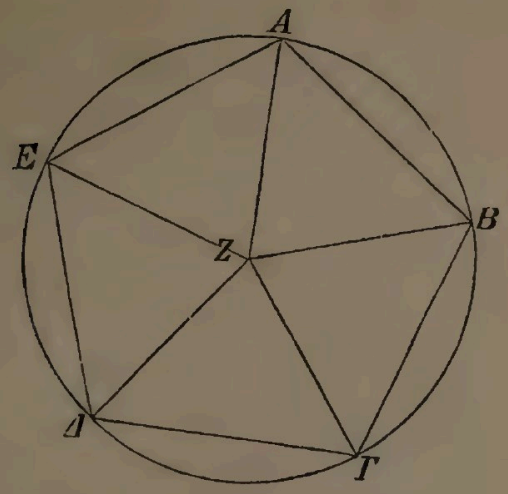
γώνου γωνίας. καὶ ἐπεὶ αἱ πρὸς τῷ Z πέντε γωνίαι τέσσαρσιν ὀρθαῖς ἴσαι εἰσὶ καί εἰσιν ἴσαι, μία ἄρα αὐτῶν, ὡς ἡ ὑπὸ AZB, μιᾶς ὀρθῆς ἐστι παρὰ πέμπτον· λοιπαὶ ἄρα αἱ ὑπὸ ZAB, ABZ μιᾶς εἰσιν ὀρθῆς καὶ πέμπτου. ἴση δὲ ἡ ὑπὸ ZAB τῇ ὑπὸ ZBΓ· καὶ ὅλη ἄρα ἡ ὑπὸ ABΓ τοῦ πενταγώνου γωνία μιᾶς ἐστιν ὀρθῆς καὶ πέμπτου· ὅπερ ἔδει δεῖξαι.

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2. εἰσί] εἰσίν PBV. 5. ZBA q. 7. ὀρθῆς ἐστι V. πέμπτον q. In fine: Εὐκλείδου στοιχείων ιγ P, Εὐκλείδου στοιχείων τῆς Θέωνος ἐκδόσεως ιγ Bq.

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Z positi quattuor rectis aequales sunt et inter se aequales, unus eorum, uelut AZB, recto angulo aequalis est deficiente quinta parte. itaque $ZAB + ABZ$



recto et quintae parti recti aequales sunt [I, 32]. et $ZAB = ZB\Gamma$. quare $AB\Gamma$ totus angulus pentagoni recto et quintae parti recti aequalis est; quod erat demonstrandum.

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